Mechanistic Interpretability: A 500-Word FAQ

What is Mechanistic Interpretability (MI)?

Mechanistic interpretability reverse-engineers neural networks to understand *how* they compute—not just which inputs matter. While post-hoc methods (gradients, saliency maps, LIME, SHAP) tell you **what** features influenced a prediction, MI reveals **the actual algorithm** the model implements internally.

Think of it like this: Post-hoc XAI is like observing that pressing the accelerator makes a car go faster. MI is like opening the hood and understanding how the engine's pistons, fuel injectors, and spark plugs work together to produce motion.

How Does MI Differ from Post-Hoc Explainability?

Post-Hoc (Gradients, SHAP, etc.)	Mechanistic Interpretability
Correlational: "These input features correlate with the output"	Causal: "This circuit *causes* the output via this computation"
Black-box: Treats model as input->output function	White-box: Examines weights, activations, attention patterns
Local: Explains single predictions	Global: Discovers general algorithms used across inputs
Approximate: Heatmaps that may be misleading	Precise: Human-readable pseudocode-level descriptions
No intervention capability	Enables steering, editing, debugging

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Core Concepts You Need to Know

1. Features & Representations

Neural networks encode information as patterns of activation across neurons. MI assumes these patterns represent human-interpretable concepts (the "Linear Representation Hypothesis"). Example: One direction in activation space might represent "mentions of deception."

2. Circuits

A circuit is a computational subgraph—a connected subset of neurons/attention heads that performs a coherent function. Famous examples: - **Induction heads**: Attention heads that implement in-context learning - **Name mover heads**: Transfer entity names across positions in sentences - **IOI circuit**: Solves "indirect object identification" tasks

3. Causal Interventions

MI doesn't just observe—it intervenes: - **Activation patching**: Swap activations between runs to test causal influence - **Ablation**: Zero out components to see what breaks - **Steering**: Modify activations to change model behavior

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Why Should You Care?

For AI Safety

MI's primary motivation isn't just scientific curiosity—it's ensuring powerful AI systems remain safe and aligned: - Detect deception or hidden objectives - Understand goal structures before deployment - Enable precise model editing without side effects

For Debugging & Improvement

- Find why models fail on edge cases
- Remove unwanted behaviors (bias, sycophancy, hallucination)
- Transfer capabilities between models

For Scientific Understanding

- Test theories about how transformers learn
- Discover universal vs. model-specific mechanisms
- Bridge ML with cognitive science and neuroscience

Key Methods at a Glance

Method	What It Does	When to Use
Probing	Train classifiers on activations to detect encoded concepts	Quick check: "Does layer N know X?"
Sparse Autoencoders (SAEs)	Decompose activations into interpretable features	Untangle polysemantic neurons
Circuit Tracing (ACDC)	Automatically discover circuits via iterative pruning	Find all components for a behavior
Logit Lens	Decode intermediate layer states to vocabulary	See what the model "thinks" mid-computation
Attention Pattern Analysis	Visualize which tokens heads attend to	Understand information flow

Common Misconceptions

"MI is just fancier feature attribution."
No—gradient methods show correlations; MI establishes causal mechanisms through interventions.

"If I can't read the weights, the model is incomprehensible."
MI has successfully reverse-engineered circuits in models up to 70B+ parameters using scalable techniques like SAEs.

"MI only works on toy models."
Recent work (2024–2026) has discovered circuits in production-scale LLMs for deception, sycophancy, and tool use.

"Understanding internals won't help with safety."
Anthropic's team uses MI to detect sleeper agents, steer refusal behavior, and audit models for hidden objectives.

Getting Started

1. **Learn transformer architecture** inside-out (attention, MLPs, residual stream)
2. **Master causal intervention techniques** (activation patching, path patching)
3. **Use tools**: TransformerLens (for small models), Neuronpedia (for SAE features)
4. **Read foundational papers**:
   - Conmy et al. (2023) on ACDC circuit discovery
   - Bereska & Gavves (2024) comprehensive review
   - Sharkey et al. (2025) open problems

The Bottom Line

Post-hoc explainability answers: *"Which inputs mattered?"*
Mechanistic interpretability answers: *"What algorithm did the model run, using which components, and how can I verify/modify it?"*

MI treats neural networks not as black boxes to approximate, but as computer programs to decompile—and it's rapidly becoming essential for anyone building, auditing, or deploying foundation models.

Resources: learnmechinterp.com (textbook) | Anthropic Interpretability | MechInterp Workshop